



C#



Overview

Introduction to C#

What is C Sharp ?

Working of C Sharp

Features of C Sharp

Basic Programming Structure

Namespaces

Data Types

Introduction

Microsoft developed C#.

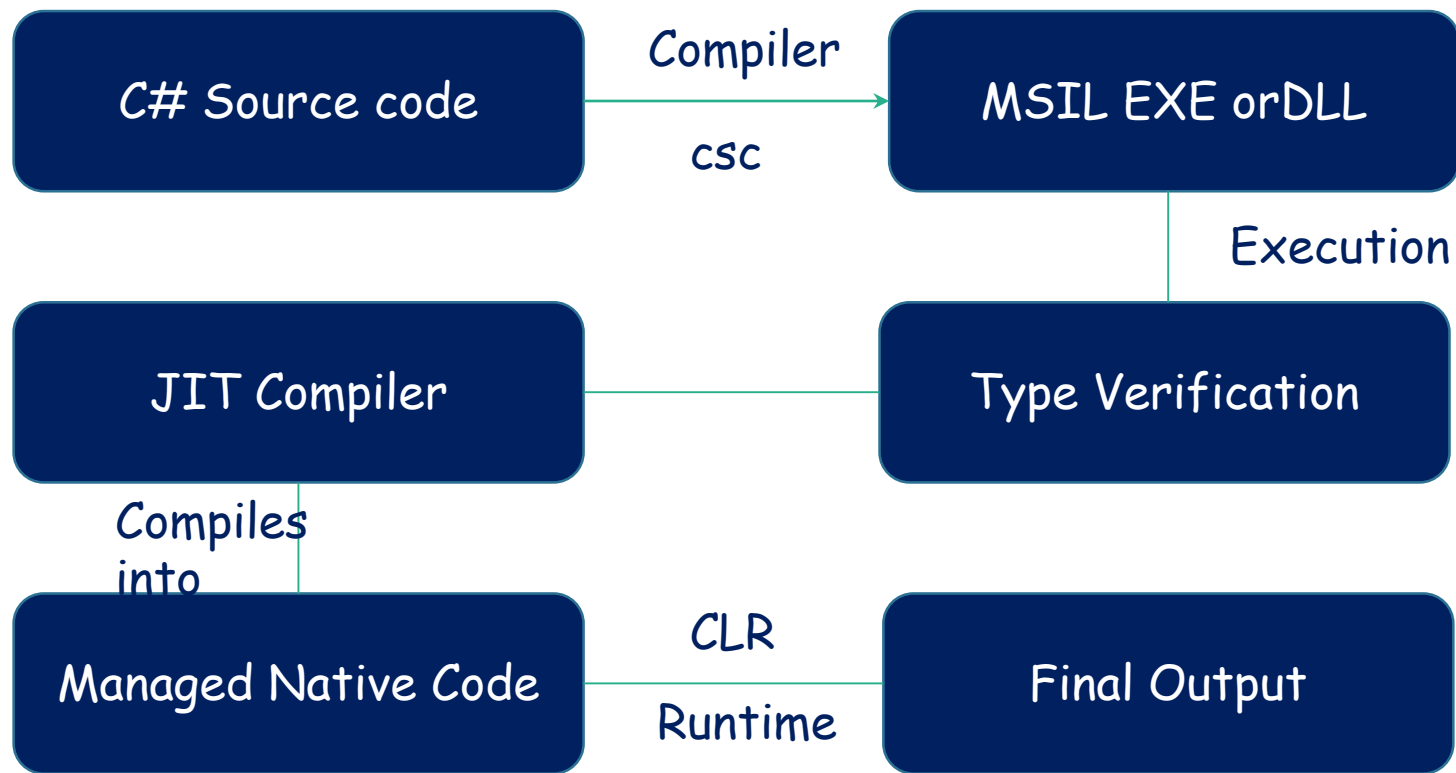
Anders Hejlsberg, the principal architect of C#.

C# will be familiar to those, who have programmed in C, C++, or Java.

c# is standardized by ECMA.



Working of C#



Compilation of C sharp

It can be compiled with the following command line:

```
csc Welcome.cs
```

This produces a file named `Welcome.exe`, which can then be executed.



What is c#

The C# programming language (pronounced "C-Sharp") .

C Sharp is a powerful object-oriented programming languages.

Microsoft developed C#, a new programming language based on the C and C++ languages.

C# is a case sensitive language.

C# support component-oriented programming.



C# Language Features

Object
Oriented.

Simple and
Flexible.

Automatic
Memory
Management.

Cross Platform
Interoperability.



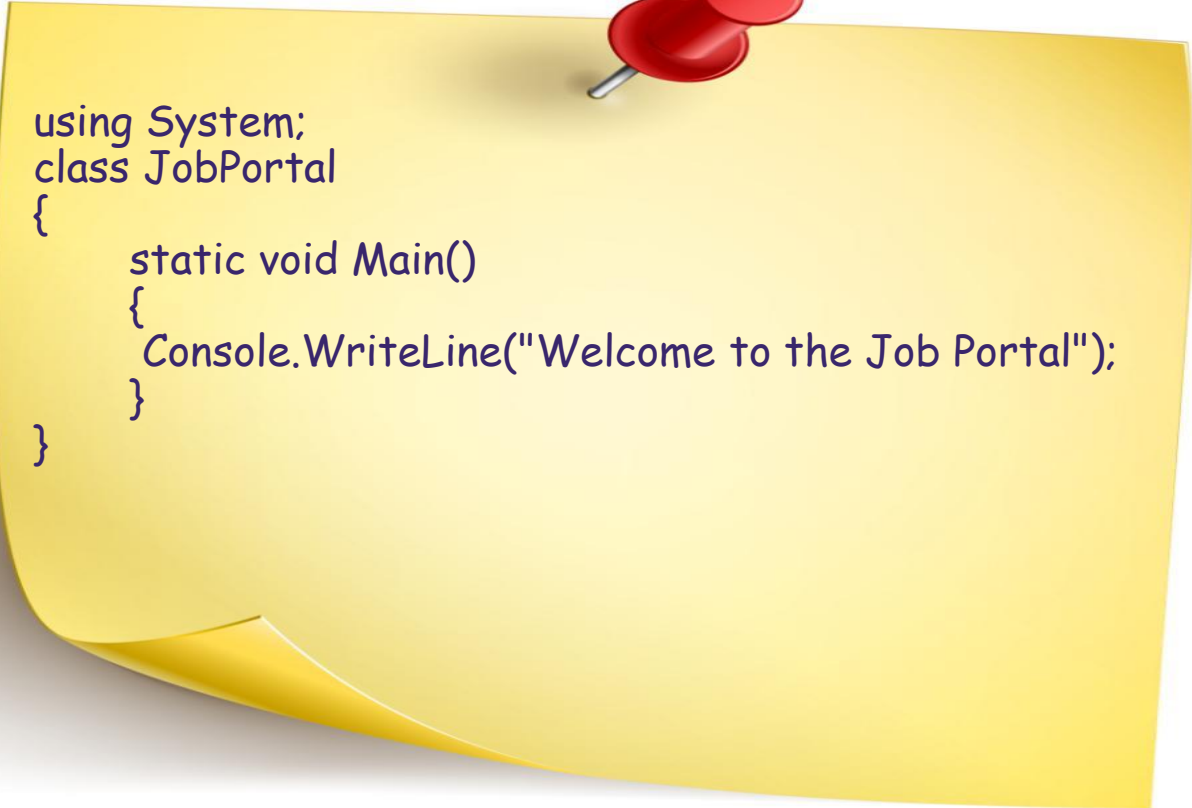
C# Editors and IDEs

Notepad

Visual Studio



Programming Structure



```
using System;  
class JobPortal  
{  
    static void Main()  
    {  
        Console.WriteLine("Welcome to the Job Portal");  
    }  
}
```



Application Types In C#

Class Library:- Creates a project for creating classes that can be used in other applications.

Console Application:- Creates a Visual C# application with a command-line interface.

Asp.Net Web Application:- Creates a Visual C# application with a Web user interface.

Asp.Net Web Service:- Creates an XML Web service with Visual C# that other applications can access.

Mobile Application:- Creates an application for mobile devices.

Programming Structure

- ❖ The- source code for a *C#* program is typically stored in one text files with a file extension of .cs
- ❖ Primary Elements
 - a namespace declaration.
 - a class.
 - a Main method.
 - a program statement.
 - a commentssay in welcome.cs



Data types

- ❑ C# supports mainly two kinds of types.
 - ✓ value types .
 - ✓ Reference types.
- ❑ value types are - char, int, structures, enums .
- ❑ Reference types are - class, interface, delegate, arrays.



Value type

A value type is a type of data in programming that directly contains its data and is usually small and simple. In simpler terms, a value type holds its value directly in memory, and when you assign it to another variable or pass it to a function, a copy of the value is made. Examples of value types include integers, floating-point numbers, characters, and small structures.

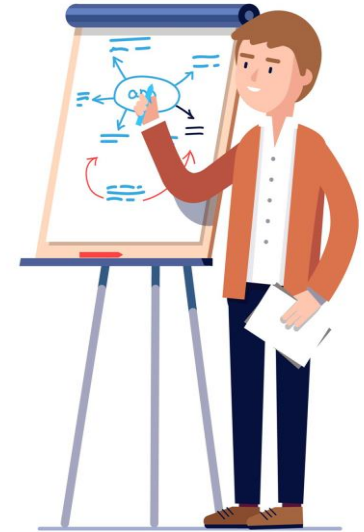


Reference types

A reference type is a type of data in programming that stores a reference, or memory address, to the location where the actual data is stored. In simple terms, when you work with a reference type, you're dealing with a pointer or a way to access the data rather than the data itself.

Unlike value types, when you assign a reference type to another variable or pass it to a function, you're passing the reference to the same piece of data, not making a new copy. This means that changes made to the data through one reference affect all references to that data.

Common examples of reference types include objects, arrays, and classes.



Conclusion

- Structure: Explored the basic structure of *C#* programs, laying the foundation for application development.
- Types: Differentiated between value types and reference types, understanding how data is handled.
- Copying: Discussed copying behavior in value types, ensuring independence between variables.
- Reference Types: Explored characteristics, including referencing instance types, default null assignment, and storing memory addresses.

These fundamentals set the stage for advanced *C#* topics ahead. Stay tuned as we delve into more complexities and practical applications.

Thank You